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TITLE PAGE

- **Problem Statement ID –** SIH26261
- **Problem Statement Title –** AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification
- **Theme –** Smart Education / Governance & Administrative Reforms
- **PS Category –** Software
- **Team ID –** [Assigned Portal Team ID]
- **Team Name (Registered on portal) –** Sentinels
- **Project Idea Title –** Aegis: Institutional Compliance Intelligence & Explainable Scrutiny Decision-Support Platform

AEGIS PLATFORM

Dual-Portal GovTech Compliance Engine for Technical Education Regulatory Scrutiny

✅ One Application Verdict Architecture (Clear / Action / Review / Incomplete)

✅ 24 Granular Forensic & Structural Checks preserved with 100% Traceability

✅ Multi-Tenant Isolation for Technical Institutions (College Portal)

✅ Zero Black-Box AI: Honest Transparency (Authenticity explicitly verified vs unverified)

✅ Sub-500ms Local Evaluation (Zero proprietary cloud API costs)

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IDEA TITLE: AEGIS

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Proposed Solution (Describe your Idea/Solution/Prototype)

🏛️ Proposed Solution Architecture

• Dual-Portal Ecosystem: Dedicated institutional workspace for Colleges (pre-submission audit) + AICTE Scrutiny Officer console.

• One Application Verdict: Replaces overwhelming 24-check lists with instant high-level verdicts: CLEAR, ACTION REQUIRED, MANUAL REVIEW REQUIRED, CHECK INCOMPLETE.

• Capabilities Matrix: Honest status for Document Reading/OCR, Compliance Rules, Cross-Document Consistency, and Authenticity Verification.

• Single 'Technical Details' Expander: Preserves 100% of forensic diagnostics, JSON evidence, and hashes behind one optional control.

• End-to-End Workflow: Supports application drafting, multi-document uploads, scrutiny queue, clarification notices, and final approval/rejection orders.

🎯 How It Addresses the Problem

• Solves Officer Cognitive Fatigue: Scrutiny officers no longer review 24 low-level technical checks per document. Clean applications are cleared in seconds.

• Pinpoints Real Exceptions: Every actionable issue shows strictly 4 fields: File, Problem, Evidence (conflicting value), and Recommended Action.

• Prevents Duplicate Evasions: Cryptographic cross-document checks detect identical PDF payloads uploaded across multiple statutory checklist requirements.

• Eliminates Scanned PDF Rejection: Classifies raster scan documents gracefully and routes them to visual seal inspection rather than causing false failures.

• Guarantees Multi-Tenant Isolation: Institution data is strictly segregated at the database query level, preventing cross-college data leaks.

💡 Innovation & Uniqueness

• Conservative Aggregation Engine: Incomplete runs, warnings, or unverified sources NEVER count as a pass. Zero false-pass guarantee.

• Honest Authenticity Truthfulness: Authenticity is explicitly marked 'Not verified' unless verified against official state issuer gateways—preventing fake AI claims.

• Non-Destructive Traceability: Aggregation is calculated non-destructively; all 24 checks, raw JSON, and execution logs remain in SQLite.

• Human-in-the-Loop Dispositions: Officers can officially record determinations (ACCEPTED, DISMISSED, NOTED) with mandatory justifications in the audit log.

• Proven Working Prototype: Fully built, verified with 34 automated unit/integration tests and 12 end-to-end browser workflows.

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TECHNICAL APPROACH

Methodology, Architecture & Implementation Pipeline

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⚙️ Core Technologies & Frameworks

Core Application Engine: Python 3.13 + Streamlit Web Framework with custom GovTech styling and role-based access control.

Document Forensics Pipeline: PyMuPDF (fitz) binary inspection, cross-reference table parsing, incremental updates (%%EOF), font streams, and active script detection (/JavaScript, /Launch).

Statutory Rules Registry: Versioned rule specifications (APH_2026_27_V1) for mandatory statutory clauses, land titles, building plans, and date plausibility.

Cross-Document Correlator: Cryptographic SHA-256 collision hashing, token set fuzzy matching for institutional entity titles, and permanent AICTE ID matching.

Relational Database & Audit: SQLite 3 with foreign key cascading, index optimization, and append-only immutable audit trail recording all regulatory events.

Automated Quality Assurance: Playwright sync Chromium engine + Pytest suite (34 automated tests, 12 browser workflows, 100% pass).

🔄 Multi-Stage Validation & Aggregation Pipeline

▶ **Stage 1: Binary & Format Gate:** Rejects 0-byte, corrupt, non-PDF, or password-locked payloads before parsing.

▶ **Stage 2: Structural Parseability:** Extracts XRef tables, detects font stream corruption, and measures page counts.

▶ **Stage 3: Forensic & Anomaly Analysis:** Flags active executable scripts, incremental save revisions, consumer PDF tool signatures (iLovePDF/Smallpdf), and inverted timestamps.

▶ **Stage 4: Statutory Compliance Rules:** Matches required APH 2026-27 clauses; gracefully classifies scanned documents to prevent false rejections.

▶ **Stage 5: Cross-Document Consistency:** Correlates payloads across checklist items, detecting duplicate files and permanent code mismatches.

▶ **Stage 6: Decision-Oriented Aggregation:** Synthesizes 24 check results into One Application Verdict (CLEAR / ACTION / REVIEW / INCOMPLETE) and displays actionable findings first.

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FEASIBILITY AND VIABILITY

Operational Feasibility, Risk Analysis & Mitigation Strategies

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📊 Analysis of Feasibility (Operational & Technical)

- **Production-Ready Architecture:** Built on standard Python and SQLite without reliance on expensive third-party AI APIs or external GPU servers.
- **Sub-Second Processing Performance:** Validates an entire 4-document dossier in under 500ms, enabling real-time feedback during applicant upload.
- **Regulatory Conformity:** Engine rules are directly mapped to statutory requirements of the AICTE Approval Process Handbook (APH 2026-27).
- **Low Total Cost of Ownership (TCO):** Lightweight footprint allows deployment on affordable government cloud servers (NIC / MeghRaj).

⚠️ Potential Challenges & Risks

- **Legacy Scanned Paper Deeds:** Historical land records and municipal approvals often exist only as raster scans without extractable digital text.
- **Consumer Editor Noise:** Applicants frequently use online PDF utilities (Smallpdf, iLovePDF) simply for file compression, triggering false forgery alarms.
- **Statutory Rule Evolution:** Regulatory handbooks and approval norms update annually, requiring dynamic rule versioning without codebase breakage.
- **Institutional Name Variations:** Minor trust name variations or accepted abbreviations (e.g., 'Tech.' vs 'Technology') causing false mismatches.

🛡️ Strategies for Overcoming Challenges

- **Adaptive Scanned Document Handling:** Classifies zero-text PDFs as raster scans; routes to 'MANUAL REVIEW REQUIRED' with seal verification prompts instead of false failure.
- **Advisory Anomaly Demotion:** Demotes consumer editor signatures and clock-drift timestamps to non-blocking advisory indicators with explicit educational officer caveats.
- **Decoupled Versioned Rule Engine:** Rule specifications (APH_2026_27_V1) are decoupled from engine logic, allowing seamless rule additions across academic years.
- **Fuzzy Token-Overlap Matching:** Employs intelligent token-set ratio matching (threshold >= 85%) ignoring standard stopwords ('Institute of', 'Society for').

IMPACT AND BENEFITS

Quantifiable Governance, Economic & Administrative Transformation

👥 Impact on Target Stakeholders

▶ AICTE Scrutiny Officers & Evaluation Committees:

- 80% Reduction in Review Time: Routine clean applications cleared in under 2 minutes instead of 30+ minutes.
- Decision Fatigue Relief: Highlights exact issues (File, Problem, Evidence, Action) rather than raw 24-check logs.
- Objective Consistency: Standardizes evaluation criteria across regional offices (Northern, Western, Southern).

▶ Technical Colleges & Higher Education Institutions:

- Real-Time Upload Guidance: Instantly notifies applicants of unreadable, password-locked, or corrupt files before final submission.
- Transparent Inquiry Loop: Official clarification notices can be reviewed and responded to with supplementary evidence directly in the portal.

▶ Council Leadership & Vigilance Authorities:

- 100% Immutable Audit Trail: Every automated check, human disposition, and approval order permanently recorded.
- Zero Tampering Window: Eliminates retroactive document swaps or undocumented manual overrides.

📈 Multi-Dimensional Benefits Matrix

🏛️ Governance & Transparency:

Enforces the 'Light but Tight' regulatory principle of NEP 2020. Eliminates human subjectivity, arbitrary delays, and corruption risks in institutional approvals.

💰 Economic & Administrative Efficiency:

Saves an estimated 40,000+ officer hours across annual approval cycles for 10,000+ technical institutions nationwide, dramatically reducing administrative overhead.

🌱 Environmental Sustainability:

100% paperless digital scrutiny process eliminates the printing, shipping, and physical archiving of millions of paper certificate copies each year.

🔒 Zero False-Pass Regulatory Security:

Conservative aggregation rules guarantee that missing documents, crashed parser checks, or unverified sources NEVER produce an accidental clean pass.

⚡ Rapid Nationwide Scalability:

Designed with modular SQLite/PostgreSQL architecture ready for immediate integration into the national AICTE Web Portal / National Single Window System.

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RESEARCH AND REFERENCES

Statutory Frameworks, Technical Standards & Verifiable Evidence

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📜 Statutory & Policy Foundations

- **AICTE Approval Process Handbook (APH 2026-27):**

Defines mandatory document checklists, land norms, and statutory certificate criteria for technical institutions.

- **AICTE Act No. 52 of 1987:**

Statutory mandate for proper planning and coordinated development of the technical education system in India.

- **National Education Policy (NEP 2020):**

Section 18: Transforming the Regulatory System into an objective, technology-enabled, and single-window evaluation framework.

- **Digital India Initiative (MeitY):**

Standards for paperless governance, e-Signatures, and automated validation under national e-Governance programs.

🔬 Technical Specifications & Standards

- **ISO 32000-1 / ISO 32000-2:**

Document Management — Portable Document Format specifications (XRef streams, %%EOF revisions, font dictionaries).

- **PyMuPDF & MuPDF Parsing Architecture:**

High-performance C/C++ native extraction engine for unencrypted streams, text extraction, and raster image detection.

- **FIPS 180-4 Secure Hash Standard:**

Cryptographic SHA-256 fingerprinting for duplicate file payload detection and audit record verification.

- **OWASP Application Security Verification:**

Protection against active malicious executable streams (/JavaScript, /Launch) and injection risks in uploaded files.

✅ Verifiable Prototype Proof

- **Full Test Coverage (34/34 Passed):**

Complete automated regression suite covering clean passes, corrupted files, active scripts, duplicate payloads, and finding dispositions.

- **Playwright Browser Scenarios (12/12):**

End-to-end verified across Public Gateway, College Representative Dashboard, Scrutiny Queue, and Review Screen E.

- **Zero False-Pass Safeguards:**

Formally proven that incomplete checks, failed rules, or missing documents never report a false CLEAR verdict.

- **Implementation Repository:**

Fully realized inside 'aegis-foundation' with complete documentation (VALIDATION_ASSESSMENT_DESIGN.md).